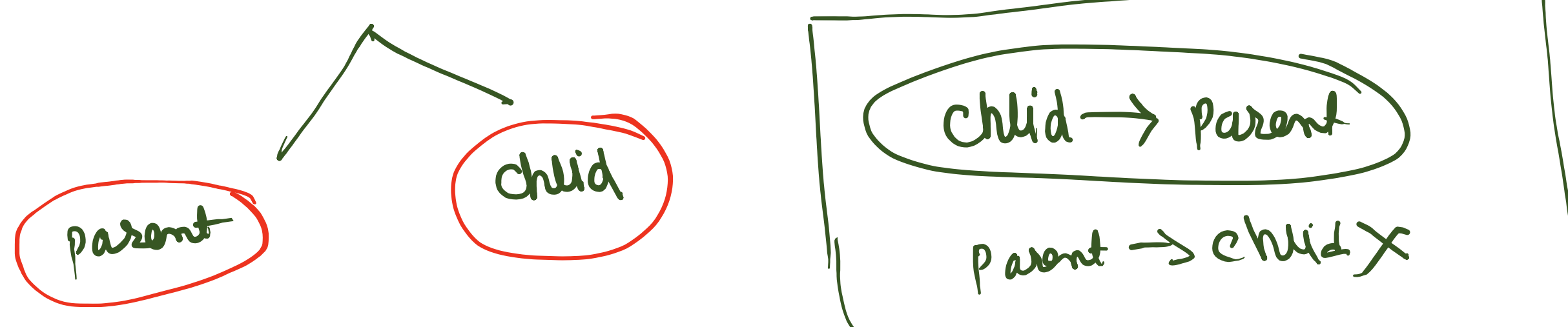
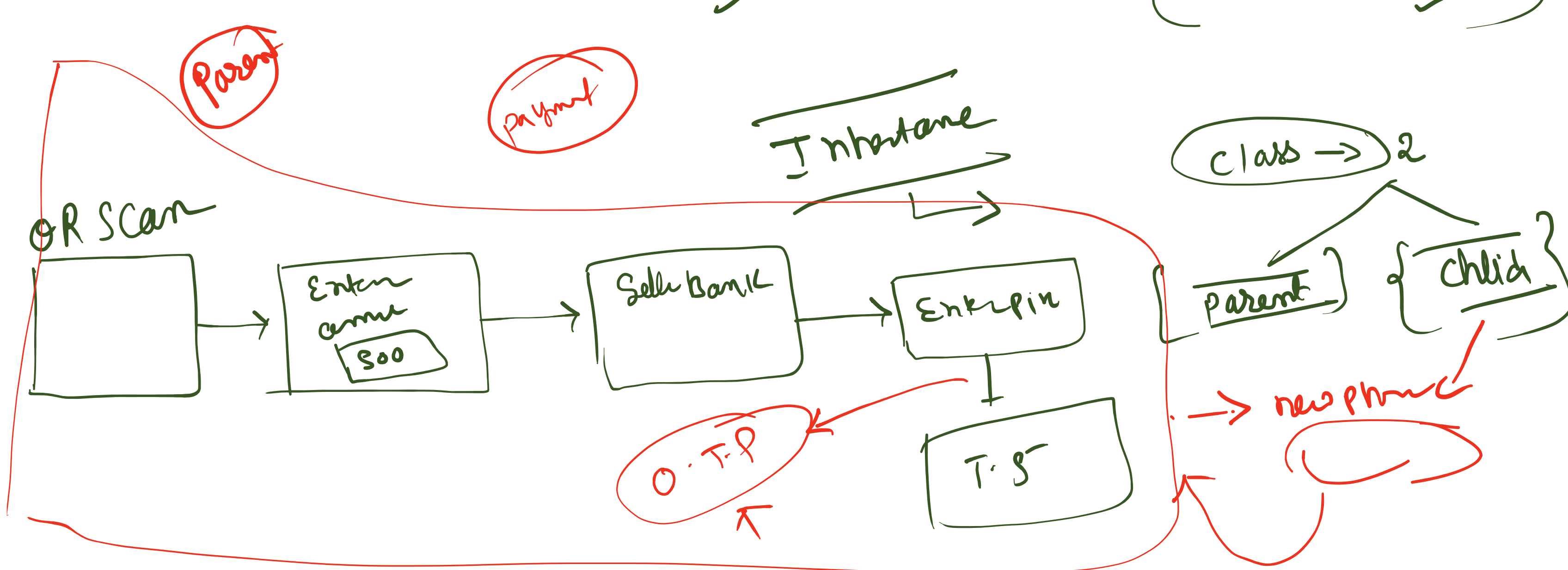


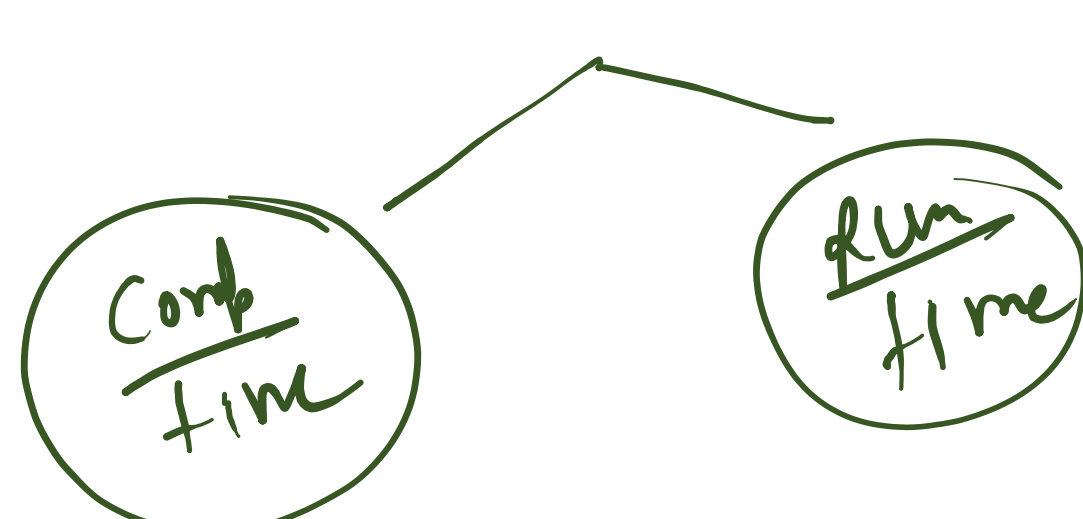
① object

Spring → content



✓ List < Integer > ll = new ArrayList<>() ✓

| ClassName | varname | = new classname() |
|-----------|---------|-------------------|
| ← (i)     | p obj   | = new p()         |
| ← (ii)    | p obj   | = new c()         |
| ← (iii)   | c obj   | = new p()         |
| ← (iv)    | c obj   | = new c()         |

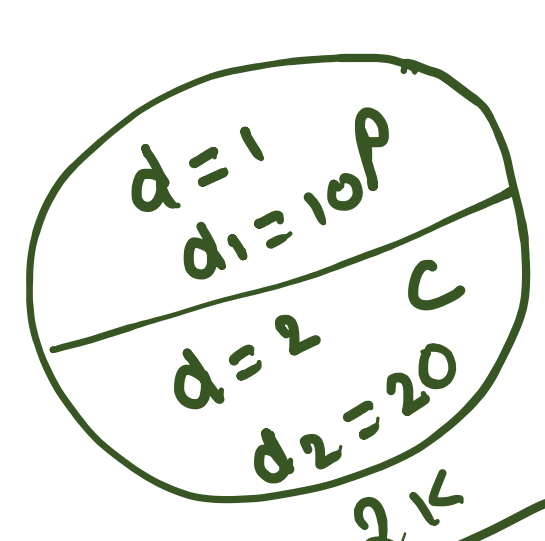
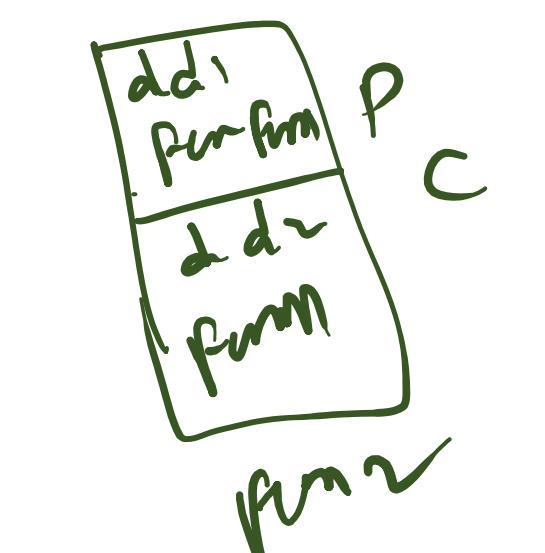


```

P obj = new C();

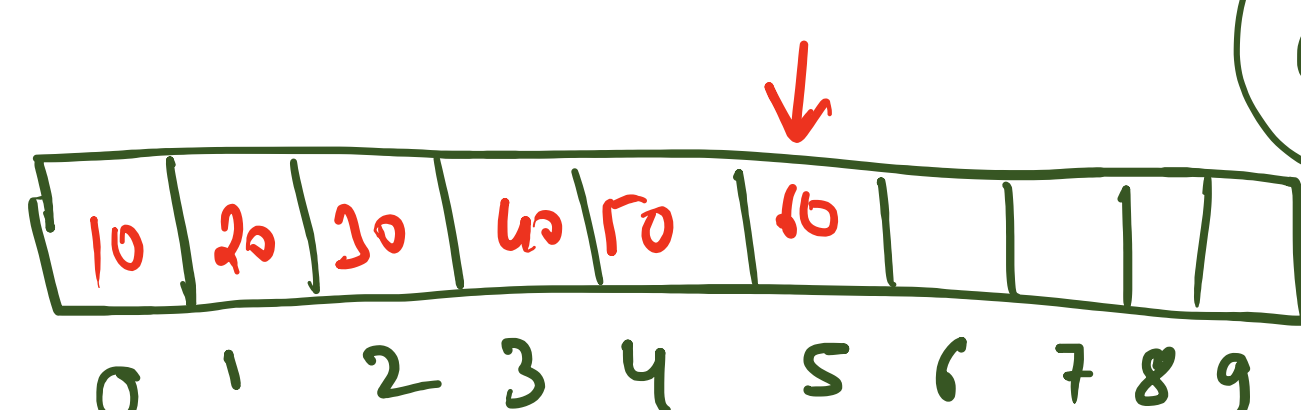
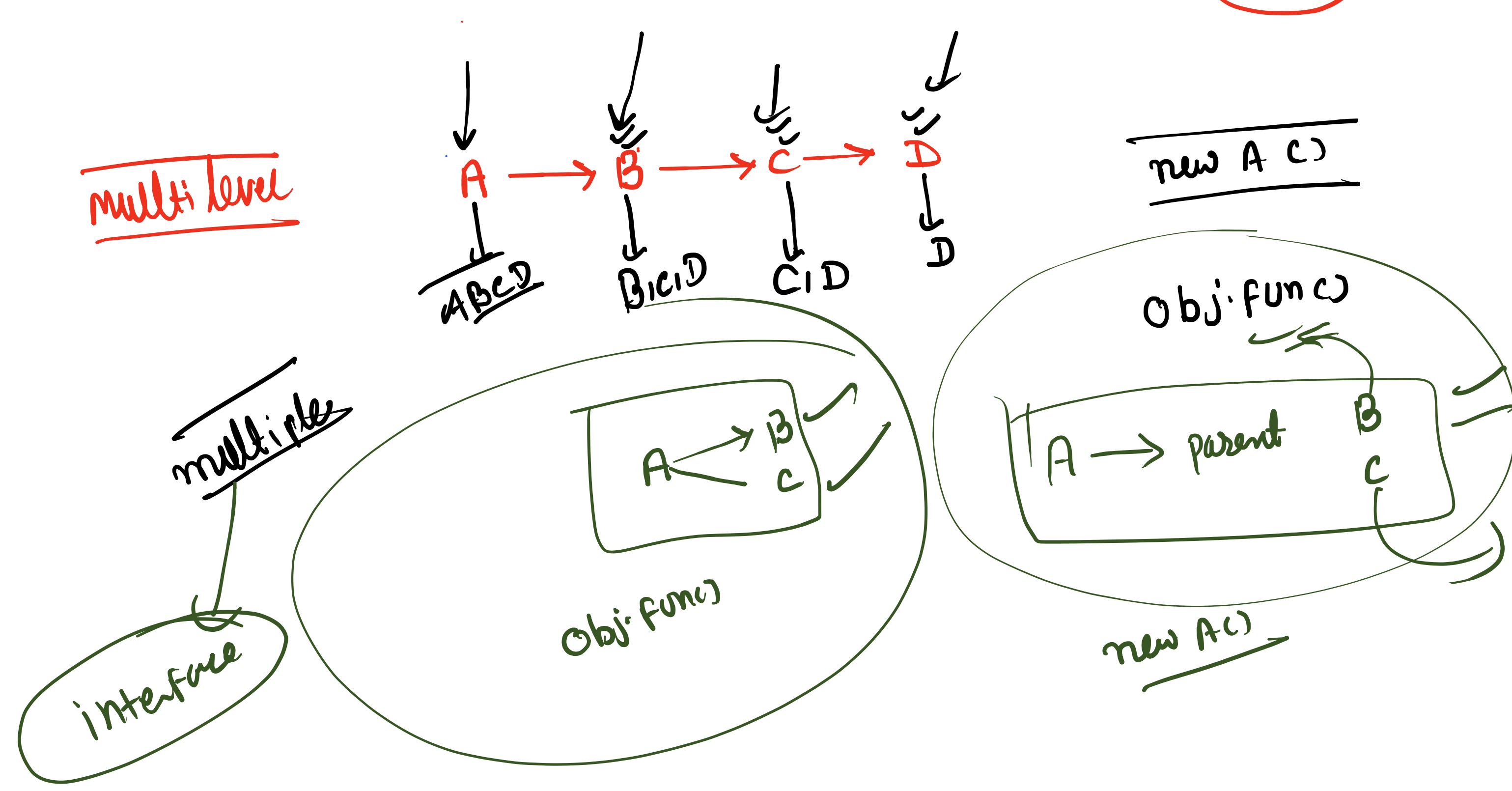
// Case - 2
P obj = new C();
System.out.println(obj.d);
System.out.println(obj.d1);
System.out.println(((C) obj).d2);
System.out.println(((C) obj).d);
obj.fun();
obj.fun1();

```

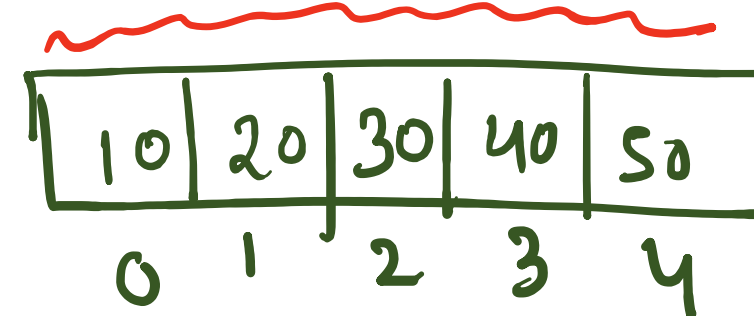


```
C obj = new P();
System.out.println(obj.d);
System.out.println(obj.d1);
System.out.println(obj.d2);
```

$d = 1$   
 $d_1 = 10$



to Spring



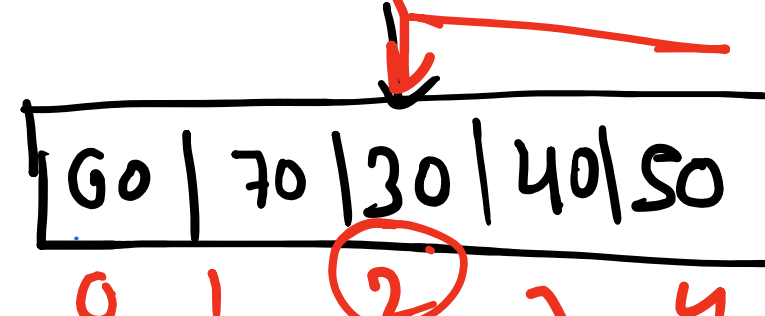
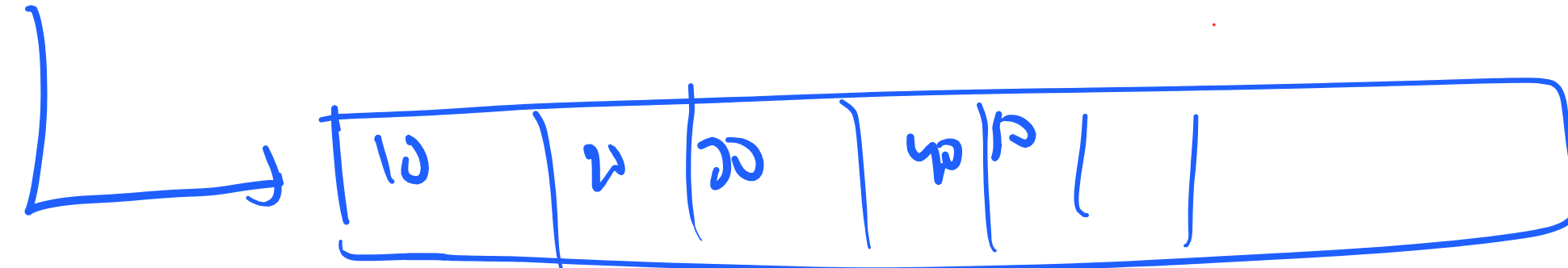
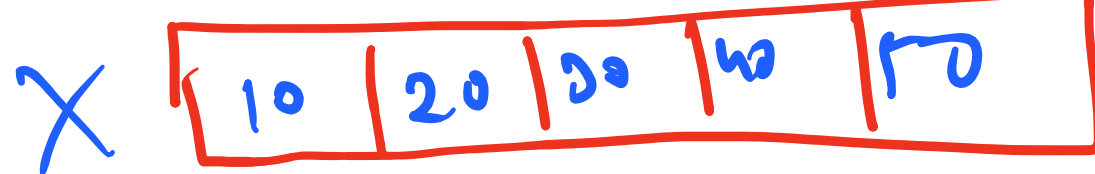
```
public class DynamicStack extends Stack {
    @Override
    public void push(int item) {
    }
}
```

```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    DynamicStack ds = new DynamicStack();  
    ds.push(10);  
    ds.push(20);  
    ds.push(30);  
    ds.push(40);  
    ds.push(50);  
    ds.push(60);  
    ds.push(70);  
    ds.display();  
}
```

```
@Override
public void push(int item) {
    if (isfull()) {
        int[] new_arr = new int[2 * arr.length];
        for (int i = 0; i < arr.length; i++) {
            new_arr[i] = arr[i];
        }
    }
}
```

```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    DynamicStack ds = new DynamicStack();  
    ds.push(10);  
    ds.push(30);  
    ds.push(40);  
    ds.push(50);  
    ds.push(60);  
    ds.push(70);  
    ds.display();  
}
```

$\text{idx} = -1$


$$\begin{aligned} \Sigma q(10) \\ \Sigma q(20) \\ \Sigma q(30) \\ \Sigma q(40) \\ \left. \begin{aligned} dq(7) \\ dq(1) \end{aligned} \right\} \\ \Sigma q(50) \\ \Sigma m(60) \\ \Sigma m(70) \end{aligned}$$
$$\begin{aligned} \text{newarr}[i] &= \text{arr}[i \times 2] \\ i &= (f + i) \% \text{arr.length} \end{aligned}$$

$i=0 \leftarrow f+0 = 2+0 = 2 \checkmark$   
 $i=1 \leftarrow \quad \quad = 2+1 = 3$   
 $i=2 \leftarrow \quad \quad \quad 2+2 = 4 \checkmark$   
 $i=3 \leftarrow \quad \quad \quad \quad 2+3 = 0 \checkmark$   
 $i=4 \leftarrow \quad \quad \quad \quad \quad 2+4 = 1 \checkmark$

$$E(8)$$